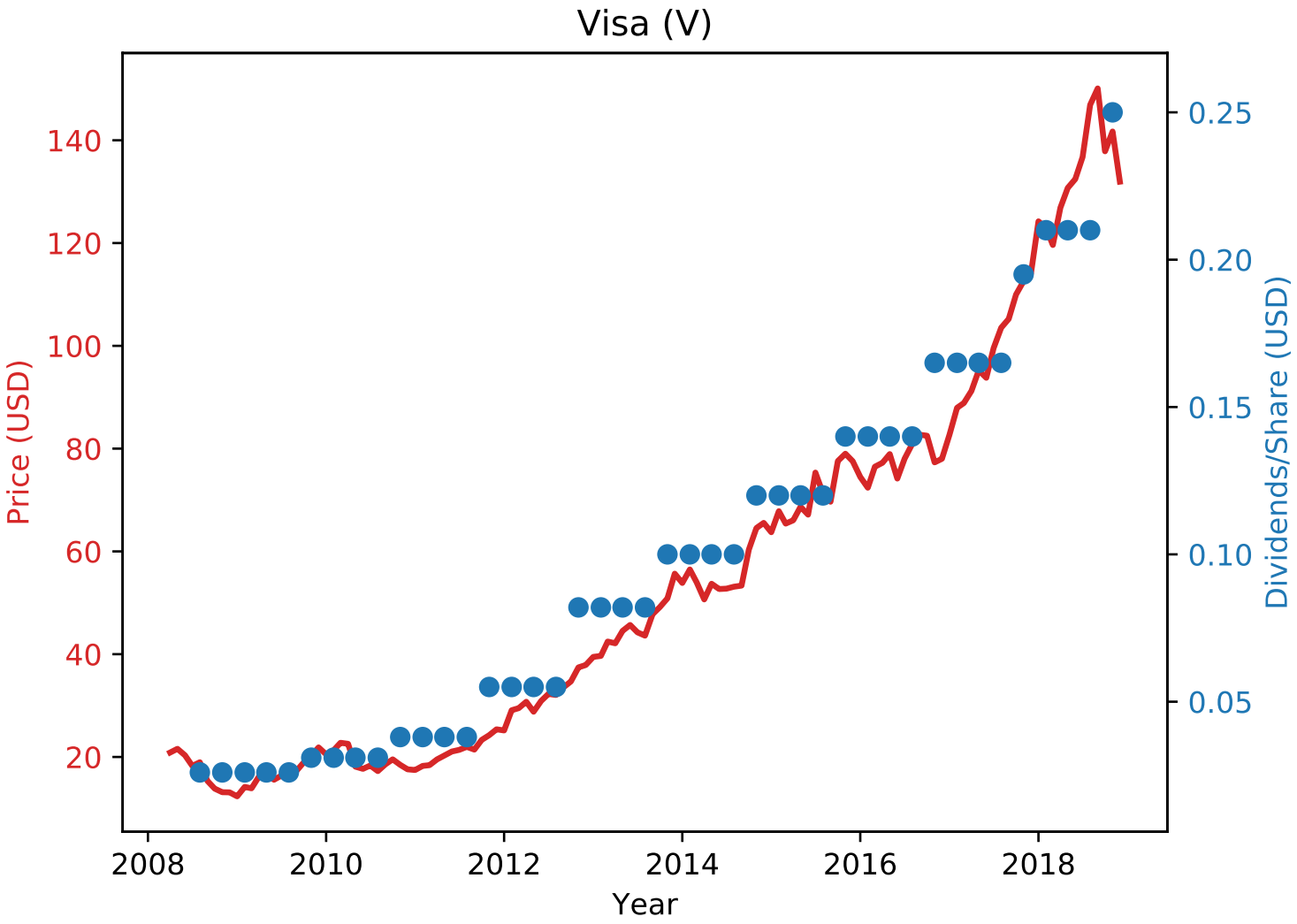
Lecture 3.3: Stock Prices, Dividends, and Returns

Stocks Prices, Dividends, and Returns

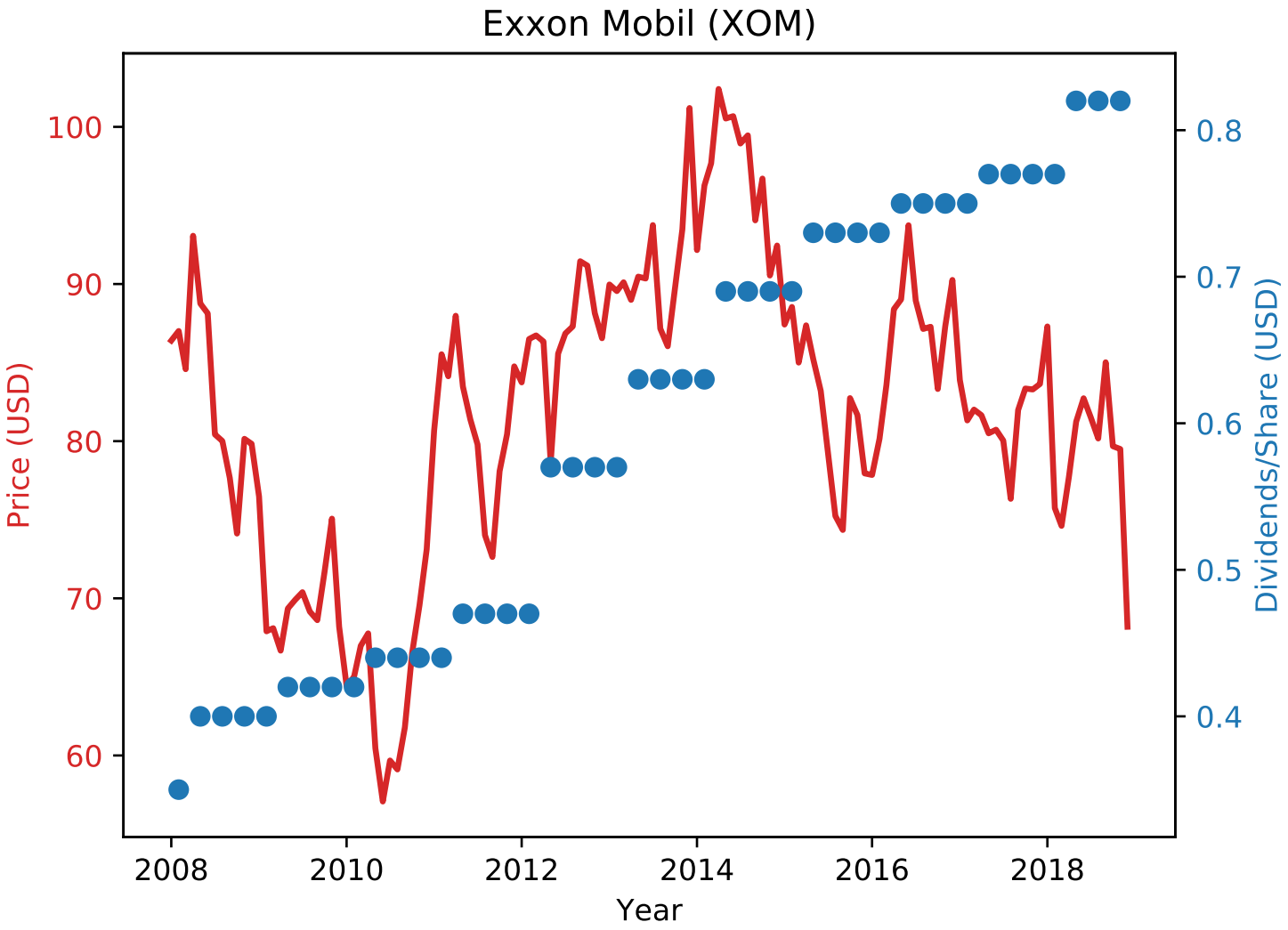
- The CAPM gives us a way of computing a stock's expected rate of return:
- We now want to price a stock's dividends using the expected rate of return from the CAPM (we call this process <u>valuation</u>)
- This is a difficult task, so we will have to make some assumptions along the way
- We will look at two methods of valuation
  1. Discounted Cash Flows (DCF) Valuation
  2. Price-to-Earnings (P/E) Valuation

Observations

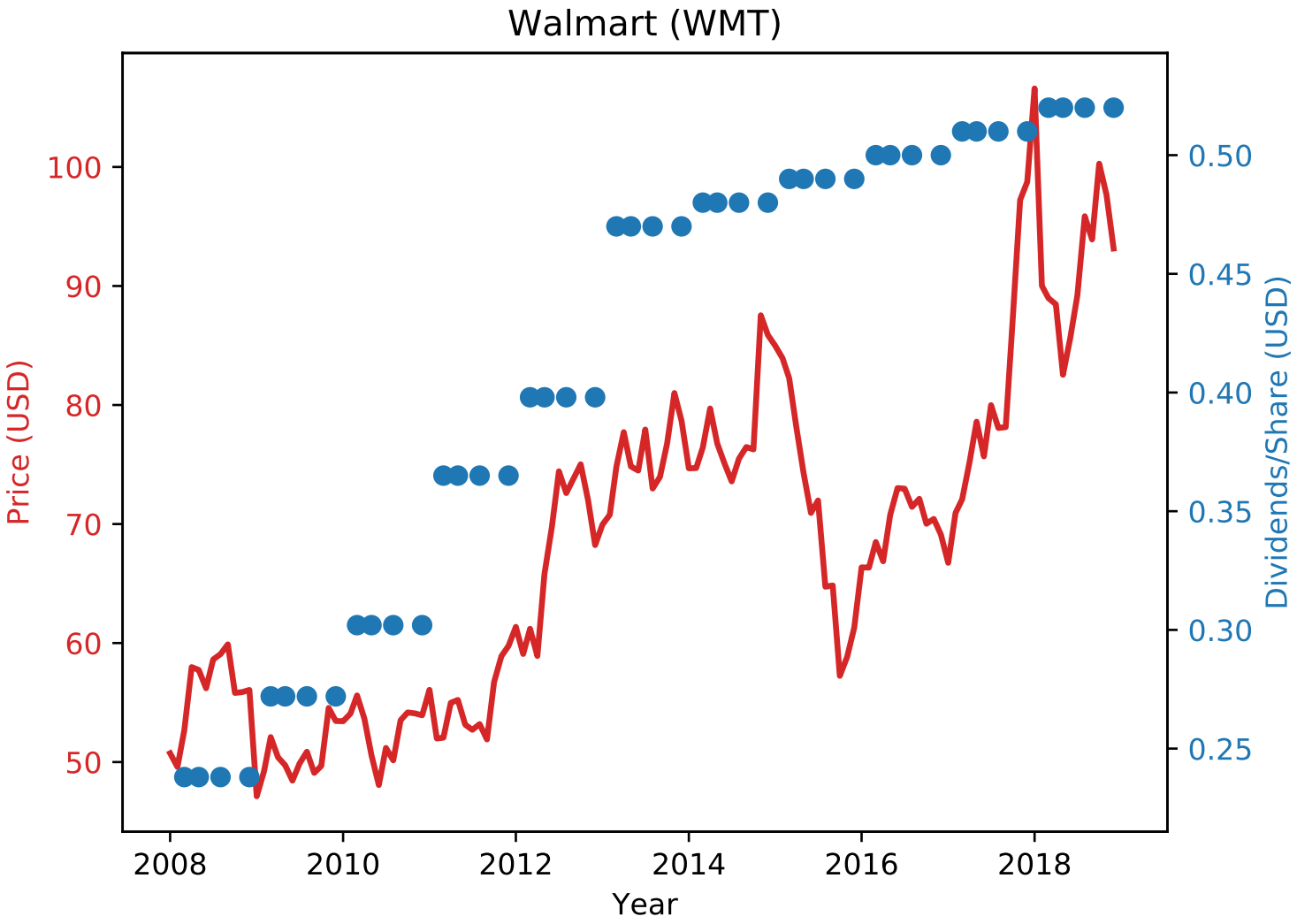
Visa


Visa (V)
Price (USD)
Dividends/Share (USD)
Year
140
120
100
80
60
40
20
0.25
0.20
0.15
0.10
0.05
2008
2010
2012
2014
2016
2018

Exxon


Exxon Mobil (XOM)
Price (USD)
Dividends/Share (USD)
Year
100
90
80
70
60
0.8
0.7
0.6
0.5
0.4
2008
2010
2012
2014
2016
2018

Walmart


Walmart (WMT)
Price (USD)
100
90
80
70
60
50
Dividends/Share (USD)
0.50
0.45
0.40
0.35
0.30
0.25
2008
2010
2012
2014
2016
2018
Year

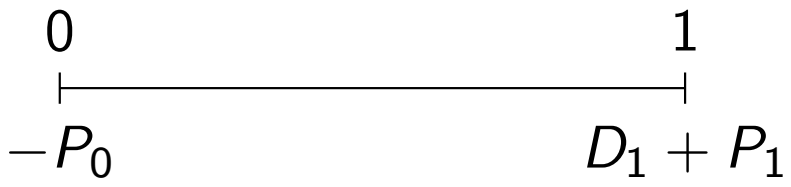
Observations

- Verizon's dividends are (generally) constant across the four quarters in each year, but increase from the fourth quarter of one year to the first quarter of the next
- Exxon's dividends follow a similar pattern to those of Verizon, despite the much greater volatility in its stock price
- Walmart's dividends follow a similar pattern to those of Verizon and Exxon, but the annual growth in the dividend rate seems to kink in 2013

DCF Valuation

Discounted Cash Flows (DCF)

- To simplify the math that follows, we will consider a stock whose dividends are uncorrelated with the market



- P_1 is the expected price in one period
- r is the required total rate of return
- D_1 is the expected dividend
- The price should be

$$P_0 \overset{\text{should}}{=} \frac{D_1 + P_1}{1 + r}.$$

Dividend Yields, Capital Gains, and Total Returns

$$r \overset{\text{should}}{=} \frac{D_1 + P_1}{P_0} - 1 = \underbrace{\frac{D_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

- The <u>dividend yield</u> is the expected annual dividend of the stock divided by its current price
- The <u>capital gain rate</u> is the percent by which the stock price increases from the current period to the next
- The <u>total return</u> (or <u>holding period return</u>) is the sum of the dividend yield and the capital gain rate

Dividend Yields, Capital Gains, and Total Returns

What are the average dividend yields, capital gains rates, and total returns for Proctor & Gamble and Apple?

Dividend Yields, Capital Gains, and Total Returns

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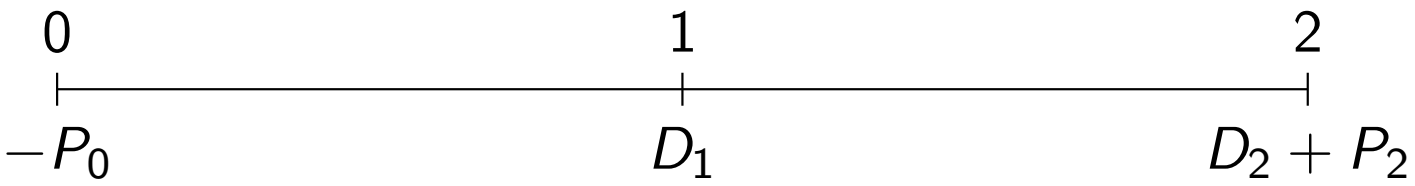
	PG	AAPL
Mean Capital Gain Rate	0.6775%	2.6681%
Mean Dividend Yield	0.3563%	0.7634%
Mean Total Return	1.0338%	3.4314%

Discounted Cash Flows (DCF)



- The prices should be

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r} + \frac{D_2 + P_2}{(1+r)^2} \text{ and } P_1 \overset{\text{should}}{=} \frac{D_2 + P_2}{1+r}.$$

- Does the two-year investor value the stock differently than the one-ear investor? No:

$$P_0 = \frac{D_1}{1+r} + \frac{D_2 + P_2}{(1+r)^2} = \frac{D_1}{1+r} + \frac{P_1}{1+r} = \frac{D_1 + P_1}{1+r}$$

Discounted Cash Flows (DCF)

- It doesn't matter whether the investor wants to hold the stock for one quarter, two quarters, or infinitely many quarters: she is always willing to pay the same price P_0
- In particular, the investor who holds the stock for infinitely many periods is willing to pay

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \cdots$$

  for the stock
- Implication: we don't need to know the prices P_1, P_2, ..., just the dividends D_1, D_2, D_3, ...

Constant Dividend Growth

Suppose dividends grow at a constant rate g

0	1	2	3	$\cdots$
$-P_0$	$(1+g)D_0$	$(1+g)^2 D_0$	$(1+g)^3 D_0$	$\cdots$

where D_0 is the dividend that was just paid

$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{(1+r)} + \frac{(1+g)^2 D_0}{(1+r)^2} + \frac{(1+g)^3 D_0}{(1+r)^3} + \cdots = \frac{(1+g)D_0}{r-g}$$

Constant Dividend Growth

- Discounted Cash Flows (DCF) says that

$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r-g} = \frac{D_1}{r-g}$$

- In this case, the <u>growth rate</u> g is the <u>capital gain rate</u>:

$$\overbrace{\frac{D_1}{P_0} + g}^{\text{Specific Formula}} \overset{\text{should}}{=} r \overset{\text{should}}{=} \overbrace{\underbrace{\frac{D_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}}^{\text{General Formula}}$$

Constant Dividend Growth

Suppose a company has an expected **quarterly** rate of return of 2%. It just paid a dividend of 0.5 USD and dividends are expected to grow by 1% each **quarter**. What would you expect the stock to trade for? What is its dividend yield? Capital gain rate?

Constant Dividend Growth

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The next dividend will be

$$D_1 = (1+g)D_0 = 1.01 \times 0.5\,\text{USD} = 0.505\,\text{USD}$$

so the price should be

$$P_0 = \frac{D_1}{r-g} = \frac{0.505\,\text{USD}}{2\% - 1\%} = 50.5\,\text{USD} \tag{1}$$

The dividend yield is $D_1/P_0 = 1\%$. The capital gain rate is the dividend growth rate: 1%.

P/E Valuation

Price-to-Earnings (P/E)

- Unfortunately, computing prices via DCF can be extremely complicated and prone to error as one must be able to forecast the firm's payout policies, how those payouts correlate with the performance of the market, and so on
- In practice, finance professionals often use multiples to price stocks (or entire companies), the most popular of which is the price-to-earnings ratio

Price-to-Earnings (P/E)

- Suppose that companies A and B have similar risk ($\beta_A \approx \beta_B$) and hence similar expected returns ($\mathrm{E}[r_A] \approx \mathrm{E}[r_B]$)
- P/E valuation assumes that earnings equal dividends: $E_A = D_A$ and $E_B = D_B$.
- Then A and B should have similar P/E ratios:

$$\frac{P_A}{E_A} \approx \frac{P_B}{E_B}$$

- If we know B's P/E ratio and A's earnings-per-share, we can approximate A's price:

$$P_A \approx E_A \times \frac{P_B}{E_B}$$

- In this case, P_B/E_B is the *multiple*

Price-to-Earnings (P/E)

Suppose that we want to price Honeywell (HON), which has a beta of 1.08. Lockheed Martin (LMT) has a beta of 0.93, while Walmart has a beta of 0.29. We should use the multiple of which company?

Price-to-Earnings (P/E)

Suppose that we want to price Honeywell (HON), which has a beta of 1.08. Lockheed Martin (LMT) has a beta of 0.93, while Walmart has a beta of 0.29. We should use the multiple of which company?

Lockheed Martin. Its beta (and hence expected return) is much closer to that of Honeywell than that of Walmart.

Price-to-Earnings (P/E)

Stock	**Beta**	**P**	**E**(PS)	**P/E**
Honeywell	1.08	$\times$	8.14 USD	$\times$
Lockheed	0.93	386.70 USD	22.82 USD	16.95 USD

What should Honeywell trade for?

Price-to-Earnings (P/E)

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Honeywell	1.08	$\times$	8.14 USD	$\times$
Lockheed	0.93	386.70 USD	22.82 USD	16.95 USD

What should Honeywell trade for?

$$
\begin{aligned}
P_{\text{HON}} &\approx E_{\text{HON}} \times \frac{P_{\text{LMT}}}{E_{\text{LMT}}} \\
&= 8.14\,\text{USD} \times \frac{386.70\,\text{USD}}{22.82\,\text{USD}} \\
&= 137.94\,\text{USD}
\end{aligned}
$$

Honeywell was *actually* trading for 161.49

Notes to self

The SDF $m(1+r_f) = 1 - \mathrm{SR}[r_m]z_m$ prices the return of the total endowment, r_m, and the risk-free rate, r_f, where $z_m = (r_m - \mathrm{E}[r_m])/\mathrm{SD}[r_m]$. So if the stock's dividends grow at a random rate of g, then

$$
\begin{aligned}
P_0 &= \sum_{k=1}^{\infty} \mathrm{E}\left[(m(1+g))^k\right] D_0 && (2)\\
&\geq \sum_{k=1}^{\infty} \mathrm{E}[(m(1+g))]^k D_0 && (3)\\
&= \sum_{k=1}^{\infty} (1+r_f)^{-k} \mathrm{E}\left[(1-\mathrm{SR}[r_m]z_m)(1+g)\right]^k D_0 && (4)\\
&= \sum_{k=1}^{\infty} (1+r_f)^{-k} \left(1+\mathrm{E}[g]-\beta_g \mathrm{E}[r_m - r_f]\right)^k D_0 && (5)\\
&= \frac{1+\mathrm{E}[g]-\beta_g \mathrm{E}[r_m - r_f]}{(r_F + \beta_g \mathrm{E}[r_m - r_f]) - \mathrm{E}[g]} D_0 && (6)\\
&= \frac{D_1 - \beta_g \mathrm{E}[r_m - r_f] D_0}{(r_F + \beta_g \mathrm{E}[r_m - r_f]) - \mathrm{E}[g]} && (7)
\end{aligned}
$$

where $\beta_g = \mathrm{Cov}[g, r_m]/\mathrm{Var}[r_m]$.